SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense

Assessment Metadata

Course Code	CSA1512
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense
Submission Type	Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed DOCX template and GitHub repository link through Google Classroom

How This Template Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product.
- Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google Classroom.

Assessment Focus

AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

Student and Capstone Details

Student Name	M.GUNA SHEKAR
Register Number	192472345
Class / Slot	Slot B

Capstone Title	GreenHire: Cloud-Based Energy-Efficient Resume-to-Job Matcher Using NLP Skill Gap Intelligence
CO1 Evidence Link	CO1 presentation and concept map demonstrating GreenHire cloud architecture and system workflow.
CO2 AT1 VM Recommendation	vCPU: _____ RAM: _____GB Storage: _____GB VM Class: _____
CO2 AT2 Evidence Link	CO2 AT2 practical implementation showing VM deployment, Docker containerization, cloud networking and storage configuration.
GitHub Repository Link	https://github.com/shekar6193/CSA0505
Submission Date	30-07-2026

Evidence Carry-Forward Summary

Evidence Source	Student Entry	How It Supports AT3 Defense
CO1 concept map / presentation	GreenHire cloud architecture, NLP workflow, user interaction diagram	Demonstrates the overall cloud solution, system components and application workflow.
CO2 AT1 sizing workbook	VM sizing recommendation (4 vCPU, 8 GB RAM, 100 GB SSD)	Justifies resource allocation and cloud infrastructure selection.
VM calibration sheet	Performance testing and VM configuration results	Confirms that the selected VM resources satisfy application performance requirements.
CO2 AT2 practical evidence	VM deployment, Docker containers, networking and storage configuration	Demonstrates successful implementation of the proposed cloud infrastructure.
Capstone architecture decision	Multi-tier containerized cloud architecture with NLP service	Supports scalability, modularity, high availability and energy-efficient cloud deployment.

Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

Design Point	Student Response
Capstone product and users	Green Hire is a cloud-based recruitment platform connecting job seekers and recruiters using NLP.
Expected workload	Thousands of resumes, job descriptions, AI-based skill matching, concurrent users.
Cloud need	Elastic computing, scalable storage, secure APIs and database hosting.
Virtualization value	Multiple isolated virtual machines improve security, flexibility and deployment.
Cost and utilization benefit	Resources are allocated only when needed, reducing infrastructure cost.
Isolation and flexibility benefit	Separate VMs for frontend, backend and database improve fault isolation and maintenance.

Case Analysis Response: Write 8-12 technical lines connecting virtualization value to your capstone product.

GreenHire is deployed on a virtualized cloud infrastructure to provide a scalable and energy-efficient recruitment platform. Virtualization enables the application to run multiple services such as the frontend, backend, NLP processing engine, and database on isolated virtual machines or containers. This separation improves security, reliability, and simplifies system maintenance. Cloud resources can be dynamically allocated based on user demand, reducing infrastructure costs and preventing resource wastage. Virtual machines allow rapid deployment, testing, backup, and recovery through snapshots and cloning. Resource right-sizing ensures efficient CPU, memory, and storage utilization while minimizing energy consumption. The cloud environment supports automatic scaling during peak recruitment periods without affecting system performance.

Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

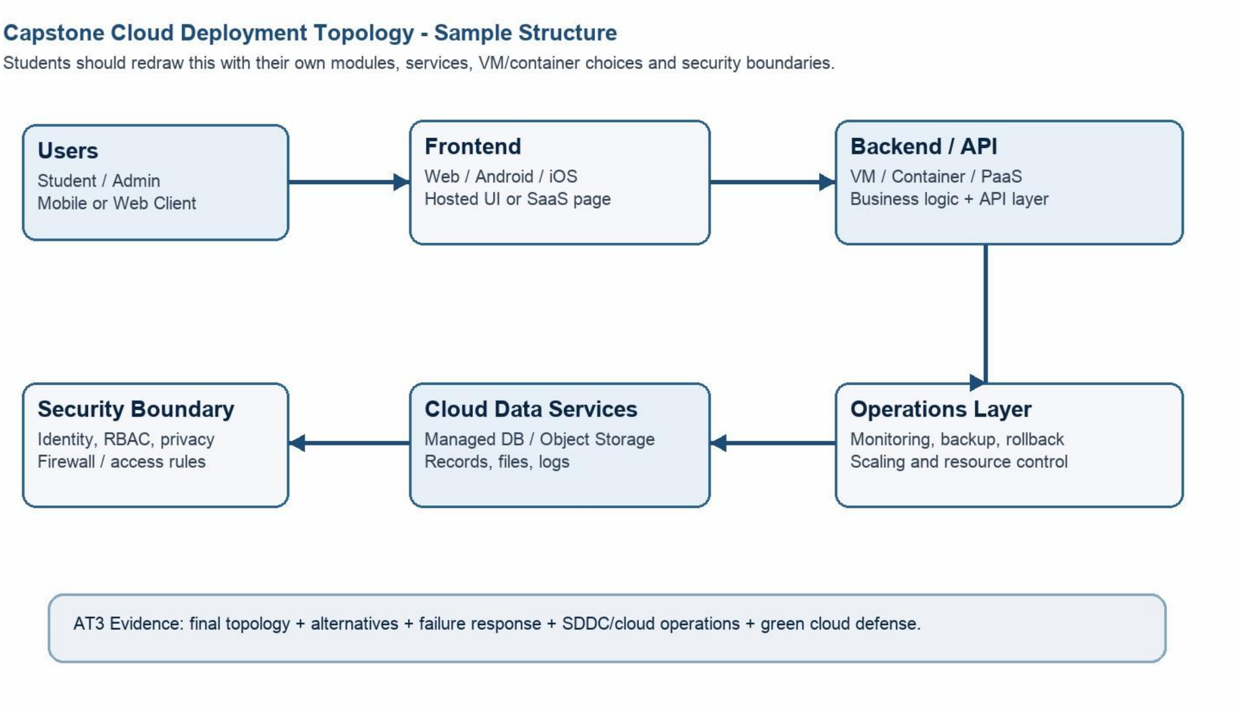
Lifecycle Element	AT1/AT2 Evidence Used	Capstone Use / Justification
VM image or base environment	Ubuntu 22.04 LTS base image with Python, Docker, PostgreSQL, and required NLP libraries installed	Provides a standardized environment for deploying the GreenHire application quickly and consistently.
Template or repeatable setup	Pre-configured VM template created during CO2 AT1	Enables rapid provisioning of identical cloud servers for development, testing, and production environments.
Snapshot / rollback	VM snapshots taken before software upgrades and configuration changes	Allows quick recovery to a stable state if deployment errors or software failures occur, minimizing downtime.
Clone / testing copy	VM clone created from the production template	Supports testing of new NLP models, APIs, and application updates without affecting the live system.
Container or VM decision	Docker containers deployed on General Purpose Virtual Machines	Containers provide lightweight deployment, faster startup, better portability, and efficient resource utilization, while VMs ensure infrastructure isolation and security.
Maintenance and update plan	Scheduled security updates, automated backups, Docker image versioning, and continuous monitoring	Keeps the infrastructure secure, improves application reliability, and minimizes service interruptions during maintenance.

Case Analysis Response: Explain how lifecycle practices reduce deployment delay and risk.

The GreenHire platform uses virtual machine lifecycle management to simplify cloud deployment and maintenance. A standardized Ubuntu VM image provides a consistent environment for hosting the application and NLP services. Pre-configured VM templates enable rapid provisioning of identical cloud instances, reducing deployment time and configuration errors. Docker containers package the frontend, backend, and NLP modules, ensuring portability and faster application startup. VM snapshots are created before updates so the system can be rolled back immediately if failures occur. VM cloning provides isolated testing environments where new features and machine learning models can be validated safely before production deployment. Automated backups, monitoring, and scheduled maintenance improve system reliability and reduce operational risk.

Q3: Final Cloud Deployment Topology and SOA/API Service Integration Case

Draw and explain the final deployment topology for your capstone product. Show frontend, backend/API, database, storage, authentication, notification, analytics and any VM/container/managed cloud service components. Explain how SOA/API-based integration connects these modules and supports modular cloud service design.



Architecture Component	Technology / Service	VM / Container / Managed Service / SaaS	Integration Method
Frontend	React.js Web Application	Docker Container	REST API over HTTPS
Backend/API	Python Flask REST API	Docker Container on Virtual Machine	RESTful APIs
Database	PostgreSQL	Managed Database Service	SQL over Secure Connection
Storage	Cloud Object Storage	Managed Cloud Storage	Cloud Storage API
Authentication	OAuth 2.0 + JWT	Managed Authentication Service	OAuth API & JWT Tokens
Notification	Email Notification Service (SMTP/SendGrid)	saas	Email api
Analytics / AI module	NLP Skill Gap Intelligence (Python, spaCy, Transformers)	Docker Container	Internal REST API

Paste or draw the final capstone deployment topology diagram here.

The GreenHire platform follows a Service-Oriented Architecture (SOA) where each major component operates as an independent service. The React frontend communicates with the Flask backend through secure REST APIs. The backend processes resume uploads and sends them to the NLP Skill Gap Intelligence module using internal APIs for skill extraction and job matching. User authentication is handled through OAuth 2.0 and JWT tokens before allowing access to application services. The backend securely stores user profiles, resumes, and job postings in the PostgreSQL database while uploaded documents are saved in cloud object storage. Notification services send interview alerts and recommendation emails through Email APIs. Because every module communicates through APIs, individual services can be updated, scaled, or maintained independently without affecting the overall system. This modular design improves scalability, maintainability, reliability, and cloud deployment flexibility.

```
        Users
          |
          ▼
   React Frontend
          |
       REST API
          |
          ▼
     Flask Backend
      /   |   \
     ▼    ▼    ▼
  NLP  Database Storage
     \    |   /
      ▼   ▼  ▼
   Authentication
          |
          ▼
     Notification
```

Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

Deployment Alternative	Advantages	Limitations / Risks	Identity and Privacy Impact
Alternative 1	Easy deployment, low initial cost, simple management	Limited scalability, single point of failure, difficult maintenance	Basic authentication with limited security and lower fault isolation
Alternative 2	Independent scaling, high availability, modular architecture, better performance	Slightly higher deployment complexity	Strong isolation, secure API communication, improved access control and privacy
Selected design	Scalable, fault tolerant, efficient resource utilization, energy efficient, easy maintenance	Requires container orchestration and monitoring	Provides secure authentication, RBAC, encrypted communication, and strong privacy protection

Security Area	Student Decision
User roles	Administrator, Recruiter, Job Seeker
Authentication method	OAuth 2.0 with JWT Token Authentication
Role-based access control	Separate permissions for Admin, Recruiter, and Candidate
Federated access / partner access	Google Sign-In and Microsoft Account Integration
Sensitive data handled	Resumes, Personal Information, Skills, Job Preferences, Recruiter Data
Privacy protection decision	HTTPS encryption, AES-256 encrypted storage, RBAC, secure APIs, regular backups, and audit logging

Case Analysis Response: Defend why the selected design is better for identity, access and privacy.

The multi-tier containerized cloud architecture is the most suitable deployment model for the GreenHire platform because it provides scalability, flexibility, and strong security. Each application component, including the frontend, backend, NLP engine, and database, is deployed independently, allowing services to be updated or scaled without affecting the rest of the system. OAuth 2.0 and JWT authentication ensure that only authorized users can access application resources. Role-Based Access Control (RBAC) restricts permissions according to user roles such as Administrator, Recruiter, and Job Seeker. Federated authentication through Google or Microsoft accounts simplifies secure login while reducing password management risks. Sensitive information such as resumes, personal details, and job data is protected using HTTPS communication, encrypted storage, secure cloud databases, and regular backups.

Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design. Include Green Computing decisions such as resource right-sizing, storage lifecycle control, log retention, API-call reduction, scheduled shutdown, VM/container/serverless comparison and carbon/energy proxy indicators where relevant.

Operational / Failure Scenario	Expected Impact	SDDC / Cloud Operations Response	Evidence or Control
VM failure	Application downtime	Restart VM and restore backup	Monitoring & Auto Recovery
Database growth	Slow performance	Increase storage and optimize database	Auto Scaling
Traffic spike	High server load	Load balancing and auto scaling	Cloud Monitoring
Credential leakage	Unauthorized access	MFA, password reset, key rotation	IAM & Audit Logs
Service outage	Temporary downtime	Backup server and failover	Disaster Recovery
Resource contention	Slow response	CPU & Memory allocation	Resource Monitoring
Cost increase	Higher cloud cost	Right-sizing and scheduled shutdown	Cost Dashboard

Operational Readiness Area	Student Plan
Automated provisioning	Infrastructure as Code (IaC)
Software-defined networking	Virtual Private Cloud (VPC)
Software-defined storage	Cloud Object Storage
Monitoring and logging	Cloud Monitoring & Logs
Backup and rollback	Daily Backup and VM Snapshots
Scaling and elasticity	Auto Scaling
Resource control and maintenance	Regular Updates and Resource Monitoring
Green Computing and sustainability control	Energy-efficient cloud resource management

Green Cloud Decision	Student Justification
VM right-sizing from CO2 AT1	Uses 4 vCPU, 8 GB RAM and 100 GB SSD to avoid resource wastage.
VM vs container vs serverless choice	Docker containers are lightweight and energy efficient.
Scheduled shutdown / idle-resource control	Automatically shuts down idle virtual machines.
Log and file retention policy	Retains logs for 30 days and archives old files.

Storage lifecycle and backup cleanup	Removes unnecessary backups and archives old data.
API-call or polling reduction	Uses caching to reduce unnecessary API requests.
Carbon/energy proxy indicator to track	Monitors CPU utilization and energy consumption.

Final Infrastructure Defense Statement: State whether the design is ready for prototype, pilot or production. Defend reliability, security, SDDC/cloud operations and Green Computing improvements required next.

The GreenHire cloud infrastructure is ready for prototype deployment. It uses virtualization, Docker containers, secure REST APIs, cloud storage, automated monitoring, backup, and auto-scaling to provide a reliable and scalable platform. Green Computing practices such as VM right-sizing, scheduled shutdown, storage lifecycle management, and API optimization improve energy efficiency and reduce operational cost. The proposed design is secure, scalable, fault tolerant, and suitable for future production deployment.

CO2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

Reflection Prompt	Student Response
What is the strongest infrastructure decision in your capstone design?	Deploying GreenHire using a multi-tier containerized cloud architecture for better scalability and reliability.
Which CO2 concept became clearer after AT1, AT2 and AT3?	Virtualization, cloud deployment, resource allocation, and SDDC concepts became much clearer.
Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability?	Privacy is the most critical risk because resumes and personal information must be protected securely.
What evidence would you show during viva to defend your infrastructure design?	VM sizing report, deployment topology, Docker deployment, cloud monitoring, and GitHub repository.
What will you improve before final capstone deployment?	Implement Kubernetes orchestration, stronger security monitoring, and AI model optimization for better performance.

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.
Virtualization and lifecycle reasoning	VM, image, snapshot, clone, container or managed-service decisions are technically justified.
Topology and integration clarity	Architecture diagram and SOA/API explanation show clear module interaction.
Identity, access and privacy	Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.
SDDC/cloud operations	Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.
Green Computing defense	Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained.
Submission quality	GitHub, template completion, evidence references and individual reflection are complete.

GitHub and Google Classroom Submission

Repository Item	Expected Content
README.md	Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index.
Completed AT3 document	This completed editable DOCX or exported PDF if instructed.
Diagram image	Final topology diagram if created separately.
Evidence references	Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.
Green Computing note	Short README section explaining sustainable cloud choices and resource-control decisions.
Google Classroom	Submit the GitHub repository link through GCR.

Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	M.GUNA SHEKAR
Register Number	192472345
Signature	M.GUNA SHEKAR
Date	30-07-2026

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	